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Subject: Workflow Automation

Assignment: BPMN Project Assignment 3

Assignment 1: Student Project Approval and Allocation

What the process is

- A team submits a final year project proposal. It gets validated, screened by the coordinator, reviewed by the committee, and then a faculty guide is matched and confirmed.
- It starts when the team is ready to submit. It ends when the project is allocated, or the proposal is closed as invalid, rejected or lapsed.

Lanes used

- One pool with 5 lanes: Student Team, Project Management System, Project Coordinator, Review Committee (incl. HoD) and Faculty Guide.
- I kept the guide as a lane inside the same pool so the guide's own task (responding to the request) can be shown.

Happy path

- Team prepares and submits the proposal.
- System validates the form and the team rules.
- Coordinator screens the proposal.
- System runs the duplicate topic check.
- Committee review (sub-process). Inside it two reviews happen at the same time, feasibility/scope and syllabus fit/novelty.
- Committee approves.
- System matches a guide by domain and workload.
- System reserves the guide slot and sends the request.
- Guide responds and accepts.
- Guide onboarding (sub-process): kick-off meeting and supervision plan.
- Record allocation and notify team & guide happen together.
- End: project allocated.

Task types and why

- User task: a person does it in the system (submit, screen, review match, late request decision, revise, guide response, manual allocation, manual notification).
- Service task: system does it automatically (validate, similarity check, reserve slot, exclude guide, record allocation).
- Business rule task: "Match guide by domain & workload", because it is decided by fixed rules like domain and max load.
- Send task: anything sent out (request to guide, rejection reasons, notification).
- Compensation task: "Release guide slot". It only runs to undo the reservation if someone withdraws.

Gateways used and why

- Exclusive gateway (X): only one path is taken. Examples: validation result, similarity above threshold, committee decision, guide found, accept team.
- Loop limits are put as conditions on the flows (attempts < 2, revision cycles < 2, attempts < 3) so loops do not go on forever.
- Parallel gateway (+): recording the allocation and notifying are not dependent on each other, so they run together and join after. Also used inside committee review for the two reviewers.
- Before the parallel join there is one exclusive merge, because notification can finish normally or through manual notification. Without it the join would get stuck.

- I did not use inclusive or event-based gateway here because there was no optional path or waiting for two events. Event-based is used in Assignment 2.

Sub-processes

- Committee review: the two reviews are inside so the main diagram is clean. The SLA timers are attached to this box.
- Guide onboarding: kick-off and signing the plan. The withdrawal message event is attached to this box.

Failure path register (Assignment 1)

F1 - Form has errors. Problem: Missing fields, no abstract, wrong roll numbers. What I did: Validation gateway sends it back to the team with the error list. Only 2 tries are allowed, after that the coordinator takes it up in screening.

F2 - Team rule broken. Problem: Team too big or a student is already in another team. What I did: Process ends with an error end event "Invalid team" and the team is told to re-form.

F3 - Duplicate topic. Problem: Similarity score above the threshold. What I did: Coordinator checks the match. If it is a false match it goes to review, if the topic needs change it goes back to submit, if plagiarised it goes to rejection.

F4 - Deadline missed. Problem: Team did not submit on time. What I did: Timer boundary event on the submit task. Coordinator decides the late request. Yes means submit again, No ends as "Proposal lapsed".

F5 - Committee asks for revision. Problem: Scope too big, objective not clear, weak novelty. What I did: Revise is allowed only while revision cycles < 2 , after that committee has to approve or reject. Revise task has a 7 day timer, if no revision it ends as lapsed.

F6 - Committee rejects. Problem: Not feasible, out of syllabus, no resources. What I did: Rejection reasons are sent. If the team has not used its new topic chance it can submit one new topic, otherwise it ends as "Proposal rejected".

F7 - Review is late. Problem: Committee does not decide within the SLA. What I did: Non-interrupting timer: at 7 days a reminder is sent, at 10 days it is escalated to HoD (escalation end event). The review still continues.

F8 - No guide available. Problem: Domain mismatch or all guides are full. What I did: Coordinator allocates manually (co-guide, external guide or waiting list) and then the slot is reserved.

F9 - Guide declines or does not reply. Problem: Busy, conflict of interest, or no reply in 2 days. What I did: Decline path and 2 day timer both go to "Exclude guide & count attempt". If attempts < 3 the rule runs again, else the coordinator allocates manually.

F10 - Notification or system fails. Problem: Email service down or database error. What I did: Notify task retries 3 times (loop). If it still fails an error boundary event sends it to the coordinator to notify manually.

F11 - Withdrawal after approval. Problem: Team breaks up or guide goes on leave. What I did: Message boundary event on guide onboarding. It throws compensation which releases the guide slot, and the team goes back to guide matching.

Assignment 2: Logistics and Shipment Exception Management

What the process is

- Shipper books a parcel. It gets picked up, moved between hubs and delivered to the recipient.
- It starts when the shipper books. It ends when the parcel is delivered and closed, returned to sender, lost, cancelled, or closed after a claim.

Lanes used

- Shipper / Customer, Control Tower / Dispatcher, Pickup Agent, Hub / Warehouse, Delivery Agent, Recipient, Claims Department.

Happy path

- Shipper creates the booking.
- Control tower validates address and weight and calculates charges.
- Booking charges are collected and pickup is scheduled.
- Pickup agent collects the parcel.
- Two things run together: parcel is scanned and goes through line-haul, and tracking is updated to the shipper.
- Line-haul (sub-process) runs once for every hub leg. Inside: sort, load & dispatch, then unload & scan at next hub.
- Both join. Not international, so customs is skipped.
- Parcel is inspected at destination hub and condition is OK.
- Recipient is told the parcel is out for delivery.
- Delivery agent delivers and takes proof of delivery.
- Shipment is closed and invoice is raised. End.

Task types and why

- User task: person does it in the system (booking, correct details, scan, supervisor scan, customs, inspection, deliver & POD, recipient address and surcharge).
- Manual task: physical work with no system (collect parcel, stop and return parcel, loading inside line-haul).
- Service task: system does it (charges, schedule pickup, reroute, hold at depot, return & adjust charges, close & invoice, declare lost).
- Business rule task: booking validation and price, because it follows fixed rules.
- Send and receive task: notifications are send tasks. For customs a send task asks for documents and a receive task waits for them.
- Compensation task: "Refund charges minus fee" undoes the charge when a booking is cancelled.

Gateways used and why

- Parallel gateway: after pickup the parcel movement and tracking update happen at the same time, then join.
- Event-based gateway: at the depot, whichever comes first, recipient collects the parcel or 7 days timer.
- Exclusive gateway: one path choices like booking valid, parcel collected, international, documents complete, parcel condition, delivery outcome, why not delivered.
- Retry limits are put as conditions on the flows (reschedules < 2, attempts < 3).
- One exclusive merge before line-haul, because normal scan, manual scan, reroute and found parcel all go back into it.

Sub-processes

- Line-haul: multi-instance, once for each hub leg. Delay, cancellation and no-scan events are attached to it.
- Trace parcel: checks last scan, manifests and vehicles. It has a 5 day timer.
- Claim: assess the claim and then refund, replace or reship.

Failure path register (Assignment 2)

E1 - Pickup failed. Problem: Shipper not there or parcel not ready. What I did: Gateway after collect. Pickup is rescheduled while reschedules < 2. On the second failure the booking is cancelled and the charge is refunded minus fee (compensation). Terminate end "Cancelled".

E2 - Wrong address. Problem: Wrong address at booking or at the door. What I did: At booking the shipper corrects it and it is checked again. At the door the recipient confirms the address. If it is outside the original zone the recipient pays a surcharge, then delivery is tried again.

E3 - Parcel damaged. Problem: Found at destination hub inspection. What I did: If insured it goes to the Claim sub-process (refund, replace or reship). If not insured, damage is recorded, shipper is informed and delivery continues.

E4 - Parcel lost. Problem: No scan for 48 hours. What I did: Timer on line-haul starts Trace parcel. If found, line-haul continues. If not found in 5 days, Claims declares it lost and pays compensation. Ends as "Closed - lost".

E5 - Delay on the way. Problem: Weather, strike, breakdown, hub full. What I did: Delay message event on line-haul. Parcel is rerouted or given another vehicle and the ETA is updated, then line-haul continues.

E6 - Customs hold. Problem: Documents missing or wrong. What I did: Documents are requested from the shipper and the process waits. If received, customs is tried again. If not received in 5 days, parcel is returned to sender.

E7 - Recipient not available. Problem: No one at the address. What I did: Delivery is tried again while attempts < 3. After the 3rd failure it is kept at the depot for 7 days. If not collected it is returned to sender.

E8 - Refused or COD fails. Problem: Recipient refuses or does not pay. What I did: Delivery outcome gateway sends it to "Return to sender & adjust charges".

E9 - Scanner or tracking failure. Problem: Scanner offline or data mismatch. What I did: Error boundary event on the pickup scan. Hub supervisor enters the scan manually and reconciles the records.

E10 - Cancelled after dispatch. Problem: Shipper cancels while parcel is moving. What I did: Message boundary event on line-haul stops it. Parcel is returned and the charge is refunded minus fee (compensation). Ends as "Cancelled".

- Terminate end events are used for cancelled and lost, because the tracking branch is still running in parallel and it also has to stop.